Ipek Naz Obek

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Education

B.A., Cognitive Science — University of Pennsylvania, Philadelphia, PA — May 2025
Concentration: Computation and Cognition
Minors: Data Science, Computational Neuroscience
Summa Cum Laude; Distinction in Cognitive Science
Benjamin Franklin Scholar; Certificate in French
Honors Thesis: *Inferring Level Progress in Goal-Directed and Open-Ended Tasks*
Advisors: Dr. Alison Gopnik, Dr. Sharon Thompson-Schill, Annya Dahmani
Relevant Coursework: Programming Languages, Probability, Data Mining, Statistical Inference, Calculus I & II, Linear Algebra, Cognitive & Computational Neuroscience, Judgment & Decisions, Behavioral Economics, Developmental Psychology

Academic Positions

Research Technician and Lab Manager — Boston University, Boston, MA — Aug 2025–present
Computation, Learning, and Development Lab
PI: Dr. Kate Nussenbaum

Research Assistant — UC Berkeley, Berkeley, CA — Jun 2024–May 2025
Gopnik Cognitive Development and Learning Lab
PI: Dr. Alison Gopnik

Honors Thesis Student — University of Pennsylvania, Philadelphia, PA — Jan 2024–May 2025
Thompson-Schill Lab
PIs: Dr. Sharon Thompson-Schill, Dr. Anna Jenkins

Research Assistant — University of Pennsylvania, Philadelphia, PA — Jun–Sep 2023
Center for Neuroengineering and Therapeutics
PI: Dr. Brian Litt

Research Experience

Computation, Learning, and Development Lab — Boston University, Boston, MA — Aug 2025–present
Research Technician and Lab Manager
PI: Dr. Kate Nussenbaum

The development of goal commitment (in progress; with Eleanor Holton): examines how goal pursuit strategies, and attention to current goals, change from childhood to adulthood.

- Adapted and programmed a goal-pursuit task (Holton et al., 2024) in jsPsych for online testing across development.
- Fit and compared computational models of alternative persistence strategies in Python; found that although children and adults performed similarly, adults persisted more with their current goals, and stronger attention to current goals predicted greater persistence.
- Analyzed trial-level choice data with mixed-effects models in R; presented at Flux 2026, manuscript in preparation.

Sunk costs and persistence (in progress; with Dr. Brian Sweis): a translational study adapting a rodent foraging paradigm to ask how time already invested shapes decisions to persist or quit.

- Designed and programmed a human version of a rodent foraging paradigm (the Web-Surf Task) in jsPsych, in which participants decide whether to wait for videos to load or skip to the next offer under a limited time budget.
- Tested participants aged 8–25 recruited through the lab's participant database and Prolific, and analyzed choice data in R.
- Identified that participants rarely quit, limiting tests of sunk-cost effects; refining the task by introducing unexpected changes in waiting time to elicit more informative quitting decisions.
- Coordinate a cross-disciplinary collaboration between the Nussenbaum and Sweis labs.

Learning progress and exploration (in progress; with Dr. Julia Leonard): an independent project testing whether exploration is guided by learning progress, by progress relative to one's expectations, or both.

- Designing an experiment in which participants predict their performance across environments that differ in learnability before choosing where to explore.
- Running model simulations to verify that the design can separate the contributions of learning progress and learning-progress prediction errors to choice.

Undergraduate research mentorship

- Mentor undergraduate Jennifer Fu on her project, *Grit as a potential moderator of susceptibility to sunk costs in decision-making tasks*, which uses the Web-Surf Task to test whether people higher in grit are more sensitive to sunk costs; meet weekly to guide her through programming the task and analyzing data.
- Mentor an undergraduate honors thesis investigating whether prior environmental volatility influences decisions to persist or give up.

Lab management

- As the lab's first lab manager, established experimental, recruitment, and data-collection infrastructure, recruiting over 1,000 participants into the lab's database for developmental studies.
- Organize a biweekly multi-lab journal club bringing together researchers to discuss current papers.

Gopnik Cognitive Development and Learning Lab UC Berkeley, Berkeley, CA Jun 2024–May 2025
Research Assistant
PI: Dr. Alison Gopnik; Advisor: Annya Dahmani

Exploration and learning progress in early childhood (with Annya Dahmani): examines how young children choose between familiar activities and new challenges, and how they track their own learning progress.

- Collected behavioral data in person at museums in San Francisco for 3 interactive virtual studies on exploration and learning progress with children aged 3–8.
- Recruited families on-site at museums every week, increasing the lab's recruitment rate.
- Wrote analysis scripts for the lab's data pipelines, including coding participants' responses as exploratory or exploitative choices.
- Designed and analyzed experiments examining how people construct and evaluate curricula for complex, multi-step tasks.
- Prepared weekly lab presentations on background literature.

Thompson-Schill Lab University of Pennsylvania, Philadelphia, PA Jan 2024–May 2025
Honors Thesis Student
PIs: Dr. Sharon Thompson-Schill, Dr. Anna Jenkins

Inferring learning progress in humans and machines (honors thesis; co-advised by Dr. Alison Gopnik and Annya Dahmani): asks how people infer how much someone has learned from their choices, and whether AI models make the same inferences.

- Conducted an extensive literature review to identify available benchmarks, datasets, and evaluation methods for measuring learning progress.
- Designed a novel behavioral paradigm in a Procgen environment (the Leaper game), in which participants saw which level a player started on and chose next, then judged how much of the first level the player had completed.
- Tested online human participants and a large language model (GPT-4o) on the same task.

- Found that people used a learner's goal to interpret ambiguous choices: when a learner repeated a level, observers who knew the goal inferred substantially more progress than those who did not, whereas GPT-4o did not consistently capture information about goals.
- Participated in weekly joint lab meetings, discussing current research and literature with faculty and lab members.

Center for Neuroengineering and Therapeutics University of Pennsylvania, Philadelphia, PA Jun–Sep 2023
Research Assistant
PI: Dr. Brian Litt; Advisor: Dr. James Gugger

Lesion characteristics and post-traumatic epilepsy (MindCORE summer fellowship): examines how the size and location of traumatic-brain-injury lesions relate to later epilepsy.

- Developed a preprocessing pipeline for clinical neuroimaging data from over 100 participants.
- Wrote Python scripts that automated labeling of lesion location and size in each brain, reducing reliance on manual annotation.
- Developed quantitative and machine-learning analyses in Python relating lesion volume and location to post-traumatic epilepsy outcomes.

Honors & Awards

Distinction in Cognitive Science University of Pennsylvania 2025
Awarded for completing an honors thesis, "Inferring Level Progress in Goal-Directed and Open-Ended Tasks," and for academic excellence in the major. Thesis advised by Alison Gopnik and Annya Dahmani (UC Berkeley Psychology) and Sharon Thompson-Schill (Penn Psychology).

Summa Cum Laude University of Pennsylvania 2025
Graduated with highest Latin honors from the College of Arts & Sciences.

Benjamin Franklin Talk Prize Benjamin Franklin Scholars, University of Pennsylvania 2025
Recognized for the best senior research presentation among the 2025 cohort of Benjamin Franklin Scholars, Penn's interdisciplinary honors program.

Lucid Communication Challenge, Engineering and Physical Sciences Talk Penn National Research Conference 2024
Selected from a field of applicants at the National Research Conference at Penn to present my honors thesis research as a talk.

Dean's List University of Pennsylvania 2021–2025
Awarded in every academic year for a GPA of 3.7 or higher.

Benjamin Franklin Scholar University of Pennsylvania 2021–2025
Member of Penn's interdisciplinary honors program in the liberal arts and sciences.

Grants & Fellowships

Pinkel Research Fund University of Pennsylvania 2025
Research funding awarded to support my senior honors thesis on curriculum learning in humans and machines.

Benjamin Franklin Research Grant Benjamin Franklin Scholars, University of Pennsylvania 2024
Summer research funding awarded by the Benjamin Franklin Scholars Program; supported my research in the Gopnik Cognitive Development and Learning Lab at UC Berkeley.

Career Services Research Grant Penn Career Services 2024
Competitive summer funding awarded by Penn Career Services; supported my research in the Gopnik Cognitive Development and Learning Lab at UC Berkeley.

MindCORE Fellowship — MindCORE, University of Pennsylvania — 2023

Selected for a fully funded 10-week summer research program in cognitive science and neuroscience at Penn's hub for the integrative study of the mind.

Undergraduate Research Mentoring Award — Penn Undergraduate Research Mentoring Program — 2023

Fellowship awarded through Penn's Undergraduate Research Mentoring Program, which funds first- and second-year students to conduct summer research with a faculty mentor.

Manuscripts in Preparation

- **Obek, I. N.**, Fu, J., Holton, E., & Nussenbaum, K. (in preparation). *Examining the development of goal commitment*.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (in preparation). *Level learning progress*.
- Fu, J., **Obek, I. N.**, & Nussenbaum, K. (in preparation). *Grit as a potential moderator of susceptibility to sunk costs in decision-making tasks*.

Unpublished Manuscripts

- **Obek, I. N.** (2025). *Inferring level progress in goal-directed and open-ended tasks*. Honors Thesis, University of Pennsylvania. Advised by A. Gopnik, S. Thompson-Schill, and A. Dahmani.
- **Obek, I. N.**, & Flombaum, J. (2021). *Are empathy and concern psychologically distinct, and is this distinction intuitive?* Research paper, Pioneer Academics Research Program.

Presentations

- **Obek, I. N.**, Fu, J., Holton, E., & Nussenbaum, K. (2026, September). *Examining the development of goal commitment* [Poster]. Flux Society for Developmental Cognitive Neuroscience, La Jolla, CA.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2025, October). *Automated curriculum learning in children and machines* [Poster]. Curiosity, Information Seeking, and Exploration Conference, Brown University, Providence, RI.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2025, October). *Automated curriculum learning in children and machines* [Talk]. Big Team Science Conference.
- **Obek, I. N.** (2025, April). *Machines that think like kids* [Talk]. Benjamin Franklin Scholars, University of Pennsylvania, Philadelphia, PA. Awarded the Benjamin Franklin Talk Prize.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, September). *Automated curriculum learning in children and machines* [Talk]. Lucid Communication Challenge, Engineering and Physical Sciences session, Penn National Research Conference, Philadelphia, PA.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, Fall). *Automated curriculum learning in children and machines* [Poster]. Center for Undergraduate Research and Fellowships (CURF) Poster Symposium, University of Pennsylvania, Philadelphia, PA.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, September). *Automated curriculum learning in children and machines* [Poster]. Penn National Research Conference, Philadelphia, PA.
- **Obek, I. N.**, Gugger, J., Narang, S., Vasireddy, M., & Zhao, C. (2023, Fall). *Understanding the link between TBI-induced lesions and epilepsy* [Poster]. Center for Undergraduate Research and Fellowships (CURF) Poster Symposium, University of Pennsylvania, Philadelphia, PA.
- **Obek, I. N.**, Gugger, J., Narang, S., Vasireddy, M., & Zhao, C. (2023, August). *Understanding the link between TBI-induced lesions and epilepsy* [Poster]. MindCORE Research Fellowship Conference, Philadelphia, PA.

Teaching

Neuroscience and Society — Teaching Assistant, 2024–2025
Instructor: Dr. Sharon Thompson-Schill — Penn Undergraduate Program in Neuroscience
Supported students through weekly office hours, in-class instruction, and group activities, and provided individualized academic guidance.

Introduction to Cognitive Science — Teaching Assistant, Fall 2023
Instructor: Dr. Russell Richie — Penn Undergraduate Program in Cognitive Science
Led weekly recitations and office hours for 20+ students, developed instructional materials, and guided students through course concepts and assignments.

MATH 114: Calculus II — Tutor, Fall 2022
Penn Weingarten Tutoring Center
Tutored first-year students for 4 hours per week, providing individualized support with mathematical problem-solving and study strategies.

Mentorship

Benjamin Franklin Scholars Mentor Program — Peer Mentor, 2024–2025
University of Pennsylvania
Mentored 7 Benjamin Franklin Scholars in their academic and social transition to Penn through regular individual outreach and community-building events.

Center for Undergraduate Research and Fellowships — Research Peer Advisor, 2024–2025
University of Pennsylvania
Advised undergraduates pursuing research opportunities and organized research programming, including an international faculty symposium on the experiences and challenges of international academics.

Pioneer Academics — Program Coordinator and Advisory Board Member 2022–2025
Advised 100+ students through the research and writing process and coordinated research seminars on effective research practices and faculty mentorship.

Young Guru Academy — Alumni Mentor 2019–present
Mentor Turkish high school and university students pursuing international education and STEM careers. Led STEM outreach across 5+ cities in rural Anatolia, training teachers and students and delivering hands-on science education to underserved communities; presented at YGA's annual summit to ~20,000 attendees.

Leadership & Service

Cognitive Science Student Association — Founder & President, 2024–2025
University of Pennsylvania
Founded and led Penn's Cognitive Science Student Association, organizing faculty talks, research seminars, mentoring, and networking initiatives connecting students with faculty, alumni, and industry professionals.

The DREAM Program — Recruitment Chair, 2023–2025
Philadelphia, PA
Led volunteer recruitment and weekly educational programming for preschool-aged children in West Philadelphia, including hands-on science activities designed to foster curiosity, collaboration, and confidence in learning.

Turkish Student Association — President, 2022–2025
University of Pennsylvania
Directed a 50-member organization and community initiatives, including raising $8K for earthquake-relief efforts.

Assembly of International Students — Social Committee Member, University of Pennsylvania — 2021–2025

Organized social and cultural programming to foster community among Penn's international student population.

The Daily Pennsylvanian — Photographer, University of Pennsylvania — 2021–2025

Photographed campus events and contributed visual content for multimedia stories at Penn's student newspaper.

Skills

Programming: Python, R, JavaScript, MATLAB, SQL, Java, OCaml

Research tools & platforms: jsPsych, PsychoPy, Qualtrics, Prolific, Pavlovia, Git/GitHub, Jupyter, LaTeX

Neuroimaging: fMRI preprocessing and GLM analysis scripts in AFNI (afni_proc.py), region-of-interest analysis, MRI lesion segmentation pipelines in Python; trained in fMRI study design and scanner operation (PSYC 4230: Introduction to fMRI)

Certifications: CITI Training (Biomedical Responsible Conduct of Research; Human Subjects Research; Social-Behavioral Research with Children)

Languages: Turkish (native), English (fluent), French (fluent)

References

Dr. Kate Nussenbaum
Assistant Professor, Department of Psychological and Brain Sciences
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Dr. Alison Gopnik
Professor of Psychology
University of California, Berkeley
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Dr. Sharon Thompson-Schill
Christopher H. Browne Distinguished Professor of Psychology; Co-Director, Center for Cognitive Neuroscience
University of Pennsylvania
sschill@psych.upenn.edu

Dr. Russell Richie
Director of Strategy and Impact
Progress and Poverty Institute
Formerly Associate Director, Cognitive Science and MindCORE, University of Pennsylvania
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